

Princeton, September 25, 1987

Dear Maxim,

Inspired by a conversation with Witten, I can now compactify - with a divisor with normal crossings at infinity - the moduli stack of super Riemann surfaces. As I have to think, I continue in French.

1. Generically Smooth Cohen-Macaulay Supercurves

In this section I propose an analogue, in algebraic supergeometry, of the notion of a geometrically reduced curve.

The accepted notion of a smooth supercurve X/S is the following: a superscheme X over the superscheme S , smooth of relative dimension $(1, 1)$, equipped with

$$\mathcal{D} \subset T_{X/S}$$

locally a direct factor of dimension $(0, 1)$, and “as non-integrable as possible.” For such a system $(X, \mathcal{D})$, one identifies $T_{X/S}/\mathcal{D}$ with $\mathcal{D}^{\otimes 2}$ by

$$D_1 \otimes D_2 \mapsto \frac{1}{2}[D_1, D_2].$$

Dualizing $\mathcal{D} \subset T_{X/S}$ gives

$$\Omega_{X/S}^1 \longrightarrow \mathcal{D}^\vee, \tag{1.1}$$

and the isomorphism $T_{X/S}/\mathcal{D} \simeq \mathcal{D}^{\otimes 2}$ dualizes to

$$0 \longrightarrow \mathcal{D}^{\vee \otimes 2} \longrightarrow \Omega_{X/S}^1 \longrightarrow \mathcal{D}^\vee \longrightarrow 0,$$

whence an isomorphism

$$\text{Ber}(\Omega_{X/S}^1) = \mathcal{D}^\vee.$$

Thus one obtains

$$\delta : \Omega_{X/S}^1 \longrightarrow \omega_{X/S} := \text{Ber}(\Omega_{X/S}^1), \tag{1.2}$$

that is, a derivation

$$\delta : \mathcal{O}_X \longrightarrow \omega_{X/S}.$$

My choice of signs may be ambiguous. For X with local coordinates (z, θ) , with

$$D = \partial_\theta + \theta \partial_z,$$

I want

$$\delta : f \mapsto ((\partial_\theta + \theta \partial_z)f)[dz/d\theta],$$

where $[dz/d\theta]$ is the basis of $\text{Ber}(\Omega_{X/S}^1)$ defined by the basis $(dz, d\theta)$ of $\Omega_{X/S}^1$.

Conversely, δ determines $\mathcal{D}$: $\mathcal{D}$ is the orthogonal of $\text{Ker}(\delta)$, or the multiples of δ in the basis of $\text{Ber}(\Omega_{X/S}^1)$. Notice that, for an epimorphism $\delta : \Omega_{X/S}^1 \rightarrow \omega_X$, it is not enough that $\text{Ker}(\delta)^\perp$ be as non-integrable as possible for δ to come from a supercurve structure: the normalization must be correct,

$$\delta f = (Df)\ell, \quad \ell = [D^2/D]^{-1}.$$

In local coordinates (z, θ) , for

$$D := \partial_\theta + \theta \partial_z$$

(an odd derivation), one has:

Lemma 1.1. For

$$\delta = (aD + bD^2)[dz/d\theta]$$

to define a supercurve structure, it is necessary and sufficient that a be invertible and that

$$[(aD + bD^2)^2 / (aD + bD^2)] = [D^2 / D],$$

i.e.

$$\text{Ber} \begin{pmatrix} a^2 + bD^2b + aDb & b \\ -(aDa + bD^2a) & a \end{pmatrix} = 1, \quad (1.1.1)$$

where

$$\text{Ber} \begin{pmatrix} a & \beta \\ \gamma & d \end{pmatrix} = (ad - \beta\gamma)d^{-2}.$$

Proof. One has

$$(aD + bD^2)^2 = (aDa + bD^2a)D + (a^2 + bD^2b + aDb)D^2.$$

The case that we shall use most is the following: $\delta = D\ell$ is a supercurve structure, $S_0 \hookrightarrow S$ is defined by an ideal of square zero, and

$$D_1 = D + a_1D + b_1D^2$$

coincides with D over S_0 . The Berezinian then reduces to a simple expression.

Corollary 1.2. Under the hypotheses just stated, the conditions of 1.1 are equivalent to

$$a_1 + Db_1 = 0. \quad (1.2.1)$$

Proof. Write

$$(2a_1 + Db_1) - a_1 = 0.$$

Of course, in the above, a and a_1 are even, while b and b_1 are odd.

Remark 1.3. (i) For X/S smooth of relative dimension $(1, 1)$, let ℓ be a basis of ω_X and let

$$\delta = D\ell : \mathcal{O} \longrightarrow \omega$$

be a derivation. The condition that D generate a local direct factor of dimension $(1, 0)$ in $T_{X/S}$ is equivalent to the surjectivity of

$$\delta : \Omega_{X/S}^1 \longrightarrow \omega_{X/S}.$$

For S an ordinary even scheme, it is also equivalent to the corresponding condition for the morphism induced by δ ,

$$\Theta_X^- \longrightarrow \omega_X^-.$$

(ii) For a surjective

$$\delta = D\ell : \Omega_{X/S}^1 \longrightarrow \omega_{X/S},$$

if on a schematically dense open set δ is a supercurve structure, then it is one everywhere: in the absence of a reference supercurve structure, the Berezinian in (1.1.1) has an invertible denominator, and since $\text{Ber} = 1$ on a schematically dense open set, the identity holds everywhere.

Definition 1.4. A supercurve over S is a superscheme X/S , flat and relatively Cohen-Macaulay, which on a fibrewise dense open set is smooth of relative dimension $(1, 1)$, equipped with a derivation

$$\delta : \mathcal{O}_X \longrightarrow \omega_{X/S},$$

where $\omega_{X/S}$ is the dualizing sheaf, such that:

(i) on each geometric fibre, δ induces an isomorphism

$$\mathcal{O}_{X_s}^- \xrightarrow{\sim} \omega_{X_s/s}^-;$$

(ii) on the smooth locus, δ is a smooth supercurve structure over S .

By 1.3(ii), condition (ii) may be replaced by:

(ii') on an open set smooth over S and schematically dense, δ defines a smooth supercurve structure over S .

1.5. For S an ordinary scheme, let us spell out this definition in terms of the scheme

$$C := (X, \mathcal{O}_X^+)$$

over S . The hypotheses of flatness, Cohen-Macaulayness, and fibrewise generically smoothness give

$$C/S \text{ is a flat curve with geometrically reduced fibres.} \quad (1.5.1)$$

The coherent sheaf

$$\mathcal{L} := \mathcal{O}_X^-$$

is flat over S , relatively Cohen-Macaulay, locally free of rank 1 on a fibrewise dense open set, and

$$\mathcal{O}_X^- \cdot \mathcal{O}_X^- = 0.$$

For such a superscheme

$$X = (C, \mathcal{O}_C \oplus \mathcal{L}),$$

the dualizing sheaf ω_X is computed as follows. We have

$$p : X \longrightarrow C,$$

and

$$\omega_{X/S} = p^! \omega_{C/S} = \mathrm{Hom}_{\mathcal{O}_C}(\mathcal{O}_X, \omega_C).$$

The relative Cohen-Macaulay hypothesis ensures the vanishing of the higher Ext's and says that $\omega_{X/S}$ is flat over S and commutes with arbitrary base change. One has

$$\omega_{X/S}^+ = \omega_C, \quad \omega_{X/S}^- = \mathrm{Hom}_{\mathcal{O}_C}(\mathcal{L}, \omega_C).$$

Moreover

$$\mathcal{O}_X^- \omega_{X/S}^+ = 0,$$

and

$$\mathcal{O}_X^- \otimes \omega_{X/S}^- \rightarrow \omega_{X/S}^+$$

is the evaluation map

$$\mathcal{L} \otimes \mathrm{Hom}(\mathcal{L}, \omega_C) \rightarrow \omega_C.$$

In the smooth case, in local coordinates (z, θ) , ω_X has basis $[dz/d\theta]$, and ω_X^- is identified with ω_C by

$$\theta[dz/d\theta] \mapsto dz.$$

A derivation

$$\delta : \mathcal{O}_X \longrightarrow \omega_X$$

is defined by

$$\delta^+ : \mathcal{O}_C \longrightarrow \omega_C \quad (\text{a derivation}),$$

and

$$\delta^- : \mathcal{L} \longrightarrow \mathrm{Hom}(\mathcal{L}, \omega_C) \quad (\mathcal{O}_C\text{-linear}).$$

The derivation conditions give the usual rules when x or y is even. For x and y odd, one should express the symmetry of the bilinear form

$$\mathcal{L} \otimes \mathcal{L} \longrightarrow \omega_C$$

defined by δ^- . Since $\mathcal{L}$ is generically of rank 1, this is automatic. Axiom 1.4(i) says that δ^- is an isomorphism. Axiom 1.4(ii) says that, on the smooth locus,

$$\delta^+ : \mathcal{O}_C \longrightarrow \omega_C$$

is

$$d : \mathcal{O}_C \longrightarrow \Omega_C^1 = \omega_C.$$

Indeed, this condition is satisfied by a smooth supercurve in standard local coordinates,

$$\theta dz = \theta[dz/d\theta] \sim dz,$$

and all δ satisfying the condition are locally isomorphic for the étale topology. Conversely, in local coordinates, if

$$\delta f = (u(z)\partial_\theta + \theta v(z)\partial_z)f [dz/d\theta],$$

then

$$(u(z)\partial_\theta + \theta v(z)\partial_z)^2 = u(z)v(z)\partial_z,$$

and the condition

$$[u(z)v(z)\partial_z/(u(z)\partial_\theta + \theta v(z)\partial_z)] = [\partial_z/\partial_\theta]$$

gives $v = 1$, as desired.

Almost: on a schematically dense open set, one has $\delta^+ = d$; hence this is true everywhere. Thus, over an ordinary scheme, giving a supercurve is equivalent to giving $C, \mathcal{L}$ as in (1.5.1), together with:

$$\text{a bilinear, automatically symmetric, form } \mathcal{L} \otimes \mathcal{L} \longrightarrow \omega_C \quad (1.5.2)$$

which induces an isomorphism

$$\mathcal{L} \xrightarrow{\sim} \text{Hom}(\mathcal{L}, \omega_C).$$

As explained, one then has

$$X = (C, \mathcal{O}_C \oplus \mathcal{L}), \quad \omega_X = \omega_C \oplus \mathcal{L}, \quad (1.5.3)$$

with the module structure defined by (1.5.2), and

$$\delta = (d, \text{id}).$$

In characteristic 2, a family of smooth supercurves is not necessarily locally, even formally, trivial. For this reason I shall henceforth assume that 2 is invertible.

So that everything up to this point could work over $\mathbf{Z}$, I used the definition of a “commutative super-ring” in which $a^2 = 0$ for odd a , and I took care to use D^2 rather than $[D, D]$ for odd derivations.

1.6. The following description of $\omega_{X/S}$ will be useful. Let

$$j : U \hookrightarrow X$$

be a fibrewise dense smooth open subset. For every section s of $j_*\omega_{U/S}$ and every connected component F of $X - U$, finite over S , a residue

$$\text{Res}_F(s) \in \mathcal{O}_S$$

is defined. The sheaf $\omega_{X/S}$ is the subsheaf of $j_*\omega_{U/S}$ consisting of those sections s such that, locally for the étale topology on S and for every such F , and every f in $\mathcal{O}_X$, one has

$$\text{Res}_F(fs) = 0.$$

Over nilpotent S , Res_F is a sum of residues along the branches passing through F and, for each branch $B \hookrightarrow X$ closed in X , is invariant under birational morphisms $B_1 \rightarrow B$. This permits one to compute ω_X in Serre’s manner.

Proposition 1.7. Let X_0 be a supercurve over a field k . An infinitesimal deformation X of X_0 that is trivial as a deformation of the underlying super-scheme is trivial as a deformation of supercurve.

Proof. It is enough to treat first-order deformations, by a Schlesinger-type argument. The case reduces to an even deformation and an odd deformation.

a. Even deformation. We use the point of view of 1.5. Suppose that X_0 is defined by C , $\mathcal{L}$, and an isomorphism

$$\mathcal{L} \xrightarrow{\sim} \mathrm{Hom}(\mathcal{L}, \omega_C).$$

A deformation with the superscheme held fixed can be viewed as a deformation of this isomorphism. Such a deformation is of the form $\varphi_0 \circ a$, where a is an infinitesimal automorphism of $\mathcal{L}$. Such an infinitesimal automorphism $a = 1 + a_1$ has square root $1 + \frac{1}{2}a_1$, and φ is conjugate to φ_0 by $(1 + a_1)^{-1/2}$. It suffices to check this on the smooth locus, where a is multiplication by an element of $\mathcal{O}_{C_0[t]/(t^2)}$ congruent to 1 modulo t . Conjugating φ does not change the isomorphism class of the deformation of supercurve, so the deformation is trivial.

b. Odd deformations. A deformation of (X_0, δ_0) , trivial as a deformation of superscheme, is of the form (X, δ) , with X obtained from X_0 by extension of scalars and

$$\delta = \delta_0 + \varepsilon D,$$

where $\delta_0 : \Omega_X^1 \rightarrow \omega_X$ is obtained from the original δ_0 by extension of scalars, and where D is an odd derivation

$$\mathcal{O}_{X_0} \longrightarrow \omega_{X_0}.$$

We must prove that, if δ satisfies 1.4(ii), then δ is conjugate to δ_0 by an automorphism of X trivial on X_0 . Such an automorphism is of the form

$$f \longmapsto f + \varepsilon E f,$$

where E is an odd derivation of $\mathcal{O}_{X_0}$ into $\mathcal{O}_{X_0}$.

From the point of view of 1.5, an odd derivation

$$D : \mathcal{O}_{X_0} \longrightarrow \omega_{X_0}$$

is given by its components

$$\begin{cases} D_0 : \mathcal{O}_C \longrightarrow \mathcal{L}, & \text{a derivation,} \\ D_1 : \mathcal{L} \longrightarrow \omega_C, & D_1(a\ell) = -D_0(a)\ell + aD_1(\ell), \end{cases}$$

where the underlined term comes from the product $\mathcal{L} \otimes \mathcal{L} \rightarrow \omega_C$.

For $\delta_0 + \varepsilon D$ satisfying 1.4(ii), D_1 is determined by D_0 . It is enough to verify this on the smooth locus. There, in local coordinates,

$$Df = (u(z)\partial_z + v(z)\theta\partial_\theta)f [dz/d\theta],$$

and

$$D_0 : f \longmapsto u(z)\partial_z f [dz/d\theta] : \Theta_{X_0}^+ \rightarrow \omega_{X_0}^-.$$

Writing

$$D = \theta v(\partial_\theta + \theta \partial_z) + u \partial_z,$$

and applying 1.2, one sees that 1.4(ii) means

$$v = \partial_z u.$$

Thus D_0 determines D .

An odd derivation E of $\mathcal{O}_X$ is described, from the point of view of 1.5, by its components

$$\begin{cases} E_0 : \mathcal{O}_C \longrightarrow \mathcal{L}, & \text{a derivation,} \\ E_1 : \mathcal{L} \longrightarrow \mathcal{O}_C, & \mathcal{O}_C\text{-linear.} \end{cases}$$

We take for E the derivation with components $(D_0, 0)$. To verify that the infinitesimal automorphism $\text{id} + \varepsilon E$ defined by E conjugates δ_0 to δ , it is enough to check this on the smooth locus, and it is enough to verify that the component 0 of $L_E(\delta_0)$ is D_0 . In local coordinates one has

$$E = u(z)\theta \partial_z, \quad \delta_0 = (\partial_\theta + \theta \partial_z)[dz/d\theta].$$

Thus

$$L_E(\delta_0) = [E, \partial_\theta + \theta \partial_z][dz/d\theta] + (\partial_\theta + \theta \partial_z)L_E([dz/d\theta]),$$

and

$$[E, \partial_\theta + \theta \partial_z] = [u(z)\theta \partial_z, \partial_\theta + \theta \partial_z] = u(z)\partial_z.$$

The second term contributes to component 1 of $L_E(\delta_0)$. We have recovered the required D_0 .

~~1.8. Nevertheless, I have finally understood what condition (ii) of Definition 1.4 means: that δ is its own transpose (up to a sign to be added).~~

2. Stable curves

Definition 2.1. A stable supercurve X/S is a proper supercurve over S , in the sense of 1.4, whose geometric fibres X_s have reductions

$$(X_s, \mathcal{O}_{X_s}^+)$$

which are stable curves.

For S the spectrum of an algebraically closed field k , it is easy, using 1.5, to determine, locally for the étale topology, the possible singularities of stable supercurves. On the singularity $xy = 0$, there are exactly two isomorphism types of coherent Cohen-Macaulay sheaves generically of rank 1. For C the curve, and B_1, B_2 its two branches, these are

$$\mathcal{O}_C \quad \text{and} \quad \mathcal{O}_{B_1} \oplus \mathcal{O}_{B_2}.$$

Both sheaves are isomorphic to their dual, with values in $\mathcal{O}_C$ or in ω_C , which is the same thing here. Moreover, every automorphism of either sheaf admits, locally for the étale topology, a square root, and 1.5 gives two possible types of supercurve singularities. Here is a description of the two possibilities by coordinates and equations. In each case I give the coordinates, the equations, a description of ω_X as a subsheaf of the direct image of its restriction to the smooth locus, and then the description of δ .

2.2. I. Coordinates z_1, z_2, θ ; equation

$$z_1 z_2 = 0.$$

The complement of the singular point is the union of two branches, with coordinates

$$\begin{cases} B_1 : z_1, \theta & (z_2 = 0), \\ B_2 : z_2, \theta & (z_1 = 0). \end{cases}$$

The sheaf ω_X is free with basis ℓ , given on the two branches by

$$\ell = \begin{cases} \left[\frac{dz_1/z_1}{d\theta} \right] & \text{on } B_1, \\ \left[-\frac{dz_2/z_2}{d\theta} \right] & \text{on } B_2. \end{cases}$$

The derivation δ is, on each branch,

$$f \mapsto \begin{cases} (\partial_\theta + \theta z_1 \partial_{z_1})f, \\ (\partial_\theta + \theta z_2 \partial_{z_2})f, \end{cases} \cdot \ell.$$

Let D denote the derivation given on the two branches by

$$\begin{cases} \partial_\theta + \theta z_1 \partial_{z_1} & (B_1), \\ \partial_\theta + \theta z_2 \partial_{z_2} & (B_2), \end{cases} \quad D : \mathcal{O} \rightarrow \mathcal{O} \text{ odd.}$$

II. Coordinates $z_1, z_2, \theta_1, \theta_2$; equations

$$z_1 z_2 = z_1 \theta_2 = \theta_1 z_2 = \theta_1 \theta_2 = 0.$$

The complement of the singular point is the union of two branches, with coordinates

$$z_1, \theta_1 \quad (z_2 = \theta_2 = 0), \quad z_2, \theta_2 \quad (z_1 = \theta_1 = 0).$$

The sheaf ω_X is generated by the following forms, given branchwise:

$$\left[\frac{dz_1}{d\theta_1} \right] := \begin{cases} \left[\frac{dz_1}{d\theta_1} \right], & \left[\frac{dz_2}{d\theta_2} \right] := \begin{cases} 0, \\ \left[\frac{dz_2}{d\theta_2} \right], \end{cases} \end{cases}$$

together with

$$\frac{\theta_1}{z_1} \left[\frac{dz_1}{d\theta_1} \right] - \frac{\theta_2}{z_2} \left[\frac{dz_2}{d\theta_2} \right] := \begin{cases} \frac{\theta_1}{z_1} \left[\frac{dz_1}{d\theta_1} \right], \\ -\frac{\theta_2}{z_2} \left[\frac{dz_2}{d\theta_2} \right]. \end{cases}$$

On each branch,

$$\delta f = (\partial_{\theta_i} + \theta_i \partial_{z_i}) f \cdot \left[\frac{dz_i}{d\theta_i} \right].$$

Set

$$\ell := \left[\frac{dz_1}{d\theta_1} \right] + \left[\frac{dz_2}{d\theta_2} \right]$$

and let D denote the odd derivation, away from the singular point, given by

$$D := \partial_{\theta_i} + \theta_i \partial_{z_i} \quad \text{on each branch.}$$

It is a rational derivation: it does not send $\mathcal{O}$ to $\mathcal{O}$. One has

$$\theta_1 \mapsto \begin{cases} 1 & \text{on } B_1, \\ 0 & \text{on } B_2, \end{cases}$$

but

$$\delta = D\ell$$

does send $\mathcal{O}$ to ω_X .

In I and II, $\omega_{X/S}$ may be computed as in 1.6. Let α_i be sections of $\omega_{X/S}$ on B_i with the singular point removed, $i = 1, 2$. For the α_i to come from a section of ω_X , it is necessary and sufficient that, for every $f \in \mathcal{O}_X$, inducing f_i on the branch B_i , one have

$$\text{Res}_0(f_1 \alpha_1) + \text{Res}_0(f_2 \alpha_2) = 0.$$

Recall that

$$\text{Res}_0 \left(\frac{\theta}{z} \left[\frac{dz}{d\theta} \right] \right) = 1.$$

In case I, each closed branch B_i may be viewed as a curve equipped with a spin structure ramified at 0: a square root of

$$\omega_{C_i}(0),$$

where $C_i = B_{i,\text{red}}$, with a simple pole allowed at 0. One glues B_1 and B_2 along the integral leaf $z_i = 0$ of the distributions

$$\partial_\theta + \theta(z_i \partial_{z_i}).$$

In case II, each closed branch B_i may be viewed as a smooth supercurve; the two branches are glued at a point.

2.3. We shall have to consider the following deformations of the supercurves I and II. The deformations are over $k[[t]]$.

I. Coordinates z_1, z_2, θ ; equation

$$z_1 z_2 = t.$$

In the plane (z_1, z_2) , the derivation $z_1 \partial_{z_1} - z_2 \partial_{z_2}$ annihilates t . The complement of the singular point $(0, 0)$ in the fibre $t = 0$ is covered by the two charts

- (1) z_1 invertible, coordinates (z_1, θ) , $z_2 = t/z_1$,
- (2) z_2 invertible, coordinates (z_2, θ) , $z_1 = t/z_2$.

The sheaf $\omega_{X/S}$ is free with basis ℓ given by

$$\ell = \frac{[dz_1 \wedge dz_2/d\theta]}{d(z_1 z_2)} = \begin{cases} \left[\frac{dz_1/z_1}{d\theta} \right] & \text{in chart (1),} \\ \left[-\frac{dz_2/z_2}{d\theta} \right] & \text{in chart (2).} \end{cases}$$

The derivation δ is given in the two charts by

- (1) $(\partial_\theta + \theta z_1 \partial_{z_1}) \ell$,
- (2) $(\partial_\theta - \theta z_2 \partial_{z_2}) \ell$.

Since δ sends $\mathcal{O}$ to $\omega_{X/S}$ away from $(0, 0)$, depth implies that this is also true at $(0, 0)$. Flatness is clear: this is a relative complete intersection.

II. Coordinates $z_1, z_2, \theta_1, \theta_2$; equations

$$z_1 z_2 = -t^2, \quad z_1 \theta_2 = t \theta_1, \quad z_2 \theta_1 = -t \theta_2, \quad \theta_1 \theta_2 = 0.$$

The charts of $X - (0, 0)$ are

- (1) z_1 invertible, coordinates z_1, θ_1 , $z_2 = -t^2/z_1$, $\theta_2 = t \theta_1/z_1$,
- (2) z_2 invertible, coordinates z_2, θ_2 , $z_1 = -t^2/z_2$, $\theta_1 = -t \theta_2/z_2$.

If A is the coordinate ring of the total space, then

$$A^+ = k[z_1, z_2, \sqrt{z_1 z_2}].$$

One may view A^+ as the subring of $k[\sqrt{z_1}, \sqrt{z_2}]$ formed by sums of monomials of integral total degree. Through

$$\theta_1 \mapsto \sqrt{z_1}, \quad \theta_2 \mapsto \sqrt{z_2},$$

one may then identify A^- with the A^+ -module of sums of monomials of half-integral, non-integral, total degree. This makes it clear that $t = \sqrt{z_1 z_2}$ is not a zero divisor, and that X is flat over $k[[t]]$.

Here are sections of $\omega_{X/S}$ away from 0, written in charts (1) and (2):

$$\begin{cases} \left[\frac{dz_1}{d\theta_1} \right], \\ -\frac{t}{z_2} \left[\frac{dz_2}{d\theta_2} \right], \end{cases} \quad \begin{cases} \frac{t}{z_1} \left[\frac{dz_1}{d\theta_1} \right], \\ \left[\frac{dz_2}{d\theta_2} \right], \end{cases} \quad \begin{cases} \frac{\theta_1}{z_1} \left[\frac{dz_1}{d\theta_1} \right], \\ -\frac{\theta_2}{z_2} \left[\frac{dz_2}{d\theta_2} \right]. \end{cases}$$

A depth argument shows that these sections extend uniquely to sections on X . They lift the generators of ω on the fibre $t = 0$. Since $\omega_{X/S}$ is flat over S and its formation is compatible with base change, they generate $\omega_{X/S}$.

Finally, δ is given in each chart by

$$\delta f = (\partial_{\theta_i} + \theta_i \partial_{z_i}) f \left[\frac{dz_i}{d\theta_i} \right].$$

The compatibility on the intersection of the charts is checked directly.

2.4. Remark. The morphism

$$\delta : \Omega_{X/S}^1 \longrightarrow \omega_{X/S}$$

is given by

$$\begin{aligned} dz_1 &\longmapsto \theta_1 \left[\frac{dz_1}{d\theta_1} \right], & d\theta_1 &\longmapsto \left[\frac{dz_1}{d\theta_1} \right], \\ dz_2 &\longmapsto \theta_2 \left[\frac{dz_2}{d\theta_2} \right], & d\theta_2 &\longmapsto \left[\frac{dz_2}{d\theta_2} \right]. \end{aligned}$$

Its image is generated by $[dz_1/d\theta_1]$ and $[dz_2/d\theta_2]$. Its cokernel has length 1, concentrated at $(0, 0)$. Indeed, if one multiplies

$$\frac{\theta_1}{z_1} \left[\frac{dz_1}{d\theta_1} \right] = -\frac{\theta_2}{z_2} \left[\frac{dz_2}{d\theta_2} \right]$$

by θ_1, θ_2, z_1 , or z_2 , one obtains respectively

$$0, \quad 0, \quad \theta_1 \left[\frac{dz_1}{d\theta_1} \right], \quad \theta_2 \left[\frac{dz_2}{d\theta_2} \right].$$

Theorem 2.5. The deformation 2.3 is the completed versal deformation of the supercurve 2.2.

More precisely, one should complete 2.2 and 2.3 at the origin and take formal versal deformations. Of course, these are deformations as supercurves, not merely as superschemes.

Theorem 2.6.

- (i) A stable supercurve has no nontrivial infinitesimal automorphism.
- (ii) The stack of stable supercurves of genus $g > 2$ is a proper and smooth algebraic stack.
- (iii) It compactifies the stack of smooth supercurves of genus g , with a normal-crossings divisor at infinity.

To tell the truth, I have verified 2.6 only modulo the super-analogue of Artin's representability theorem and the super-analogue of Schlessinger's formal prorepresentability theorem.

Standard arguments reduce 2.6(i),(ii) to 2.6(i) and 2.5, together with verification of the valuative criterion of properness. For the latter, it is enough to consider only schemes as bases and to pass to the language of 1.5. Finally, to prove 2.5, it suffices to prove it to first order: if 2.3 over $k[t]/(t^2)$ is the only first-order deformation, then the tangent space to the versal deformation space has dimension $(1, 0)$, and every deformation over $k[[t]]$ extending the one over $k[t]/(t^2)$ is versal.

It was by examining the valuative criterion of properness that I came upon the deformations 2.3. From this point of view they are not surprising. The need for type II comes, for example, from the fact that if a curve of genus g degenerates into two curves of genera g_1 and g_2 attached at one point, then ω_{X_0} has no square root, being of odd degree on each component.

The divisors at infinity are characterized by the fact that, on each of them, one of the singular points admits, locally for the étale topology, an equation 2.2. Remark 2.4 provides, over each of them, a section, “the singular point”, whose image is locally the support of

$$\text{coker}(\delta : \Omega_{X/S}^1 \longrightarrow \omega_{X/S}).$$

These divisors at infinity are subvarieties of codimension $(1, 0)$.

I confess that for 2.5 I have only a disgusting proof: the sheaf of calculations that follows. It is somewhat rough, but in writing it up cleanly I would hardly do better.

3. First-Order Deformations

3.1. For the moment, let us forget δ and compute the versal deformation, to first order, of the singularity 2.2, I or II. The computation is painful, but standard. For an arbitrary superscheme with an isolated singularity, one must:

(a) write

$$X = \text{Spec}(A), \quad A = k[X, \Theta]/\langle f_i \rangle;$$

(b) find a system of syzygies S_k ,

$$\sum_i a_i(X, \Theta) f_i = 0,$$

generating all of them modulo the trivial syzygies

$$f_i f_j - f_j f_i = 0 \quad (\text{even, or even/odd, case}),$$

$$f_i f_j + f_j f_i = 0 \quad (f_i \text{ and } f_j \text{ odd});$$

(c) compute the cohomology of the complex of free A -modules

$$\bigoplus_i \left(A \frac{\partial}{\partial X_i} \oplus A \frac{\partial}{\partial \Theta_i} \right) \xrightarrow{df/\partial X} \bigoplus_i A \frac{\partial}{\partial f_i} \xrightarrow{\text{coefficients}} \bigoplus_k A \frac{\partial}{\partial S_k}.$$

The H^1 parametrizes the deformations.

Case I. The coordinates are (z_1, z_2, θ) , the equation is

$$z_1 z_2 = 0,$$

and there is no syzygy. The universal first-order deformation is

$$z_1 z_2 = t + \tau \theta \quad (\text{over } k[t, \tau]/(t^2, t\tau)).$$

Case II is another matter.

Case II. We take the coordinates

$$z_1, z_2, \theta_1, \theta_2$$

and the equations

$$z_1 z_2 = z_1 \theta_2 = \theta_1 z_2 = \theta_1 \theta_2 = 0.$$

The syzygies are

$$\begin{cases} \theta_1(z_1 z_2) = z_1(\theta_1 z_2), \\ \theta_2(z_1 z_2) = z_2(z_1 \theta_2), \\ z_1(\theta_1 \theta_2) = \theta_1(z_1 \theta_2), \\ z_2(\theta_1 \theta_2) = -\theta_2(\theta_1 z_2). \end{cases}$$

One has to consider a system of four equations, one per syzygy, in four unknowns, one per equation, in the ring A , expressing that these unknowns

satisfy the syzygies. The task is to compute the space of solutions modulo the trivial solutions, obtained by differentiating the equations. I denote the unknowns by

$$++, \quad +-, \quad -+, \quad --.$$

The equations corresponding to the syzygies have coefficients

	++	+-	-+	--		∂_{z_1}	∂_{z_2}	∂_{θ_1}	∂_{θ_2}
1	θ_1	0	$-z_1$	0	++	z_2	z_1	0	0
2	θ_2	$-z_2$	0	0	+-	θ_2	0	0	z_1
3	0	$-\theta_1$	0	z_1	-+	0	θ_1	z_2	0
4	0	0	θ_2	z_2	--	0	0	θ_2	$-\theta_1$

each row, one syzygy, each column, one trivial solution.

Let (a, b, c, d) be a solution. Modifying it by the last two trivial solutions, one may suppose that d is even. The equations

$$z_1 d = \theta_1 b, \quad z_2 d = -\theta_2 c$$

then give $d = 0$. The linear combinations with $d = 0$ of the trivial solutions are the linear combinations of the first, the second, the third multiplied by θ_2 , and the fourth multiplied by θ_1 . This reduces us to the system

	++	+-	-+	-- = 0		1	2	$3\theta_2$	$4\theta_1$
1	θ_1	0	$-z_1$	0	++	z_2	z_1	0	0
2	θ_2	$-z_2$	0	0	+-	θ_2	0	0	$\theta_1 z_1$
3	0	$-\theta_1$	0	0	-+	0	θ_1	$\theta_2 z_2$	0
4	0	0	θ_2	0	--	0	0	0	0

Let (a, b, c) be a solution, with $d = 0$. The equation

$$\theta_1 a = z_1 c$$

imposes that, modulo (θ_2, z_2) , the pair (a, c) is a multiple of the second trivial solution. In particular

$$a \in (z_1, \theta_1, z_2, \theta_2) = (z_1, \theta_1) \oplus (z_2, \theta_2).$$

Similarly for b and c :

$$a = a_1 + a_2, \quad b = b_1 + b_2, \quad c = c_1 + c_2.$$

Modulo the first two trivial solutions, one obtains

$$a = 0, \quad b_2 = 0, \quad c_1 = 0.$$

By the last equations,

$$a_1 \in (\theta_1), \quad a_2 \in (\theta_2), \quad b_2 = c_1 = 0,$$

$$a_1 \in (\theta_1), \quad a_2 \in (\theta_2), \quad b_1 \in (\theta_1), \quad c_1 = 0, \quad b_2 = 0, \quad c_2 \in (\theta_2).$$

Finally, modulo the θ -multiples of the first trivial solutions, every solution is reduced to one of the following:

	a	b	c	d
1	θ_1	0	0	0
2	θ_2	0	0	0
3	0	θ_1	0	0
4	0	0	θ_2	0

Thus the versal first-order deformation is given by the parameters

$$t_1, t_2, \tau_1, \tau_2,$$

the coordinates

$$z_1, z_2, \theta_1, \theta_2,$$

and the equations

$$\begin{cases} z_1 z_2 + \tau_1 \theta_1 + \tau_2 \theta_2 = 0, \\ z_1 \theta_2 + t_1 \theta_1 = 0, \\ z_2 \theta_1 + t_2 \theta_2 = 0, \\ \theta_1 \theta_2 = 0. \end{cases}$$

3.2. Singular Deformations

According to 2.7, the versal moduli space of deformations of a singularity of a supercurve is a subspace of the versal moduli space of the underlying superscheme. To first order, one has the following result.

Lemma 3.3. The versal moduli space of a singularity of a supercurve is, to first order, the following subspace of the versal moduli space, computed in 3.1, of the superscheme singularity:

$$\begin{aligned} \text{I: } & \tau = 0, \\ \text{II: } & \tau_1 = \tau_2 = 0, \quad t_1 + t_2 = 0. \end{aligned}$$

The point is to know when one can deform δ , while respecting the following conditions:

- (a) δ takes values in ω_X ;
- (b) on the smooth locus, δ satisfies 1.4(ii).

In the deformations under consideration, parametrized by a vector subspace of the one that parametrizes the deformation of 3.1, the complement U of the singular point is the union of two branches. On these branches we take the respective coordinates (z_1, θ_1) and (z_2, θ_2) . We describe a section ω of ω_X by its restrictions ω_1 and ω_2 to the two branches. The condition on ω_1 and ω_2 is that, for every section f of $\mathcal{O}_X$, with restrictions f_1 and f_2 to the two branches, one has in $\mathcal{O}_S$

$$\text{Res}_0(f_1\omega_1) + \text{Res}_0(f_2\omega_2) = 0.$$

Since ω_X is flat over S , to obtain a set of generators of ω_X it is enough to lift a set of generators of ω_{X_0} , where X_0 is the special fibre.

In case I, respectively II, let b denote the following section of ω_X on U :

$$\begin{aligned} \text{I: } & \begin{cases} (1) \left[\frac{dz_1/z_1}{d\theta} \right], \\ (2) \left[-\frac{dz_2/z_2}{d\theta} \right], \end{cases} \\ \text{II: } & \begin{cases} (1) \left[\frac{dz_1}{d\theta_1} \right], \\ (2) \left[\frac{dz_2}{d\theta_2} \right]. \end{cases} \end{aligned}$$

Let D_0 be the odd derivation on U :

$$\begin{aligned} \text{I: } & \begin{cases} (1) \partial_\theta + \theta \partial_{z_1}, \\ (2) \partial_\theta - \theta \partial_{z_2}, \end{cases} \\ \text{II: } & \begin{cases} (1) \partial_{\theta_1} + \theta_1 \partial_{z_1}, \\ (2) \partial_{\theta_2} + \theta_2 \partial_{z_2}. \end{cases} \end{aligned}$$

Set $\delta_0 = D_0 \cdot b$. This δ_0 does not necessarily send $\mathcal{O}_X$ to ω_X ; for a deformation as a supercurve to be possible, one must be able to correct δ_0 so that it sends $\mathcal{O}_X$ to ω_X .

We distinguish cases I and II, and the even and odd deformations.

Case I, odd. The deformation is

$$z_1 z_2 = \tau \theta.$$

Here b lies in ω_X ; one has

$$b = \frac{[dz_1 dz_2 / d\theta]}{d(z_1 z_2 - \tau \theta)},$$

so it is a basis of ω_X . One would have to correct the odd derivation D_0 , that is, pass to $D_0 + \tau D$, where D is an even derivation with values in $\mathcal{O}$. Describing a section of $\mathcal{O}_X$ by its restrictions to the branches, one has

$$z_1 = \begin{cases} (1) & z_1, \\ (2) & \tau \theta / z_2 \end{cases} \xrightarrow{D_0} \begin{cases} \theta z_1, \\ \tau / z_2, \end{cases}$$

which is not in $\mathcal{O}_X$: θz_1 is in $\mathcal{O}_X$, while τ / z_2 is not, since 0 is not in $(1/z_2)$. The derivation D would have to send z_1 to a nonzero quantity on the second branch; this is impossible.

Case II. Absence of odd deformations. The X_0 is bihomogeneous; one can assign the weights

$$\theta_1 : (1, 0), \quad z_1 : (2, 0), \quad \theta_2 : (0, 1), \quad z_2 : (0, 2).$$

The versal deformation of the singularity is also bihomogeneous:

$$\tau_1 : (1, 2), \quad \tau_2 : (2, 1), \quad t_1, t_2 : (1, 1).$$

Therefore, if there is a deformation with

$$\tau_1 = a\tau, \quad \tau_2 = b\tau, \quad (a, b) \neq (0, 0),$$

there is also one with (a, b) replaced by $(\lambda a, \mu b)$. If a or b is zero, then by symmetry there is a deformation with $\tau_2 = 0$; otherwise, there are two distinct deformations and τ_1, τ_2 are in fact free.

It is therefore enough to study the deformation over $k[\tau_1]$:

$$z_1 z_2 = \tau_1 \theta_1, \quad z_1 \theta_2 = z_2 \theta_1 = \theta_1 \theta_2 = 0.$$

We compute ω_X . Write an element of ω_X according to its values on each branch:

$$\begin{pmatrix} u_1 \left[\frac{dz_1}{d\theta_1} \right] \\ u_2 \left[\frac{dz_2}{d\theta_2} \right] \end{pmatrix}.$$

One has to lift the generators. The generator $[dz_1/d\theta_1]$ is not, as it stands, in ω_X , because

$$z_2 \left[\frac{dz_1}{d\theta_1} \right] = \begin{cases} \frac{\tau_1 \theta_1}{z_1} \left[\frac{dz_1}{d\theta_1} \right], \\ z_2 \cdot 0, \end{cases}$$

has residue τ_1 . It lifts to

$$\begin{cases} \left[\frac{dz_1}{d\theta_1} \right], \\ -\frac{\tau_1 \theta_2}{z_2} \left[\frac{dz_2}{d\theta_2} \right]. \end{cases}$$

Similarly, $[dz_2/d\theta_2]$ lifts to

$$\begin{cases} 0, \\ \left[\frac{dz_2}{d\theta_2} \right]. \end{cases}$$

Finally,

$$\frac{\theta_1}{z_1} \left[\frac{dz_1}{d\theta_1} \right] - \frac{\theta_2}{z_2} \left[\frac{dz_2}{d\theta_2} \right]$$

lifts by the same expression on both branches.

Now seek to lift

$$\delta = D_0 \cdot b$$

as in 3.1, II. The rational lift given by the same formula on each branch is not valued in ω_X : one has

$$\theta_1 = \begin{cases} \theta_1, \\ 0 \end{cases} \mapsto \left[\frac{dz_1}{d\theta_1} \right].$$

The derivation D_0 would have to be corrected by an even rational derivation sending

$$\theta_1 \mapsto \frac{\theta_2}{z_2} \left[\frac{dz_2}{d\theta_2} \right]$$

plus a regular section on the second branch, which is impossible.

Case II, even deformations. One has

$$\begin{aligned} z_1 z_2 &= 0, & \theta_1 \theta_2 &= 0, \\ z_1 \theta_2 &= t_1 \theta_1, & z_2 \theta_1 &= t_2 \theta_2. \end{aligned}$$

On the two branches:

branch 1: $z_2 = 0, \quad \theta_2 = t_1 \theta_1 / z_1, \quad \text{coordinates } (z_1, \theta_1),$

branch 2: $z_1 = 0, \quad \theta_1 = t_2 \theta_2 / z_2, \quad \text{coordinates } (z_2, \theta_2).$

We again look for generators of ω_X , in the form

$$\begin{cases} u_1(z_1, \theta_1) \left[\frac{dz_1}{d\theta_1} \right], \\ u_2(z_2, \theta_2) \left[\frac{dz_2}{d\theta_2} \right], \end{cases} \quad u_i \in k[z_i, z_i^{-1}; \theta_i].$$

The generator $[dz_1/d\theta_1]$ lifts as

$$\begin{pmatrix} \left[\frac{dz_1}{d\theta_1} \right] \\ -\frac{t_1}{z_2} \left[\frac{dz_2}{d\theta_2} \right] \end{pmatrix},$$

and $[dz_2/d\theta_2]$ lifts as

$$\begin{pmatrix} -\frac{t_2}{z_1} \left[\frac{dz_1}{d\theta_1} \right] \\ \left[\frac{dz_2}{d\theta_2} \right] \end{pmatrix}.$$

Moreover,

$$\begin{pmatrix} \frac{\theta_1}{z_1} \left[\frac{dz_1}{d\theta_1} \right] \\ -\frac{\theta_2}{z_2} \left[\frac{dz_2}{d\theta_2} \right] \end{pmatrix}$$

lifts in the same way.

Let D_0 be as above. One has

$$D_0(z_1) = \begin{pmatrix} \theta_1 \left[\frac{dz_1}{d\theta_1} \right] \\ 0 \end{pmatrix}, \quad D_0(z_2) = \begin{pmatrix} 0 \\ \theta_2 \left[\frac{dz_2}{d\theta_2} \right] \end{pmatrix},$$

which are in ω_X . Next

$$D_0(\theta_1) = \begin{pmatrix} \left[\frac{dz_1}{d\theta_1} \right] \\ \frac{t_2}{z_2} \left[\frac{dz_2}{d\theta_2} \right] \end{pmatrix}, \quad \text{is in } \omega_X \text{ if } t_1 = -t_2,$$

and

$$D_0(\theta_2) = \begin{pmatrix} \frac{t_1}{z_1} \left[\frac{dz_1}{d\theta_1} \right] \\ \left[\frac{dz_2}{d\theta_2} \right] \end{pmatrix}, \quad \text{is in } \omega_X \text{ if } t_1 = -t_2.$$

If $t_1 + t_2 \neq 0$, one must correct D_0 to $D_0 + \varepsilon D$, with D even; once again, D would have to send θ_1 to a nonzero quantity on the second branch, which is impossible.

Thus one can deform only when

$$t_1 + t_2 = 0.$$

That the non-excluded deformations are possible follows from 2.5.

4. Infinitesimal automorphisms: proof of 2.6(i)

A stable supercurve has no non-trivial infinitesimal automorphism.

Locally, on the smooth locus, the sheaf $\mathcal{A}$ of infinitesimal automorphisms projects isomorphically onto $T_X/\mathcal{D}$, where $\mathcal{D}$ denotes the structural distribution. For

$$X = (C, \mathcal{O}_C \oplus \omega_C^{1/2}), \quad \mathcal{A} = \omega_C^{-1} \oplus \omega_C^{1/2} \otimes \omega_C^{-1}.$$

The first task, near a singularity, is to compute which meromorphic sections of $\mathcal{A}$ are infinitesimal automorphisms. One has to send Θ into Θ .

Case I. Here Θ_X^- is a square root of ω_C . Let D be as in 3.1, I, where it was denoted D_0 . An infinitesimal automorphism is written as

$$aD + bD^2 \quad \text{with} \quad (-1)^{\deg a} 2a + Db = 0,$$

or equivalently

$$a = \mp \frac{1}{2} (-1)^{\deg b} Db.$$

In order for it to send Θ into Θ , one needs:

$$\begin{aligned} \text{even derivation : } & b \text{ even, } b \in \mathcal{O}^+, \\ \text{odd derivation : } & b \text{ odd, } b \in \mathcal{O}^-. \end{aligned}$$

Thus, in a neighbourhood of a component C ,

$$\boxed{\mathcal{A} = \omega_C^{-1} \oplus \omega_C^{1/2} \otimes \omega_C^{-1}}.$$

1

Case II. Similarly, take D as in 3.1, II, and write the derivations as

$$\begin{aligned} \text{even : } & \frac{1}{2} Db \cdot D + bD^2 \quad (b \text{ even}), \\ \text{odd : } & -\frac{1}{2} Db \cdot D + bD^2 \quad (b \text{ odd}). \end{aligned}$$

The condition for b , a rational section of $\mathcal{O}$, to give rise to a derivation $\Theta \rightarrow \Theta$, is

$$\begin{aligned} \text{even : } & b \in \mathcal{O}^+, \quad b \text{ vanishes at } 0, \\ \text{odd : } & b \in \mathcal{O}^-, \quad b \text{ vanishes at } 0. \end{aligned}$$

In terms of the branches B_i , one has

$$\mathcal{A} = \mathcal{A}_1 \oplus \mathcal{A}_2, \quad \mathcal{A}_i = T_{B_i}(-0) \oplus T_{B_i}(-0) \otimes \omega_{B_i}^{1/2}.$$

¹On the smooth locus, $[D, X + aD] \bmod \mathcal{D} = DX + (-1)^{|a|} a \partial_z$; thus there is a unique a that brings the bracket back into $\mathcal{D}$.

An automorphism $\sim \text{id}$, even or odd, therefore gives, on the normalization of a component A of C , a non-zero section of a line bundle of degree

$$\begin{aligned} \text{even : } & (2 - 2g) - \frac{1}{2}(\# \text{ points of type I}) - (\# \text{ points of type II}), \\ \text{odd : } & (1 - 2g) - \frac{1}{2}(\# \text{ points of type I}) - (\# \text{ points of type II}). \end{aligned}$$

Under the stability hypothesis, these degrees are strictly negative. Hence there is no non-trivial infinitesimal automorphism.

5. Valuative criterion of properness

Here the only parameter spaces one has to consider are traits (S, η, s) , with $S = \text{Spec}(V)$, and one can therefore use the point of view of 1.5. By Deligne–Mumford, the ordinary stable curve C_η degenerates uniquely into a stable curve C over S , after ramifying S . What remains, after a further ramification if necessary, is to make L_η and the pairing with values in $\omega_{C/S}$ degenerate.

Away from the singular points, one can extend L_η to an invertible sheaf L . Correcting L by a divisor concentrated on the special fibre, and after passing to a double cover of S , one may assume that

$$L^{\otimes 2} \longrightarrow \omega_{C/S}$$

is an isomorphism away from the singular points. After ramification, the unique extension of this L will do.

In an étale neighbourhood of each singular point, the square roots of ω_X on the complement of the singular point are classified by

$$H^1(\mu_2) = H^1(\mathbf{Z}/2).$$

Choose a section s . In L , over $X - s$, the $\sqrt{z}$'s form a double cover. The variety of vanishing cycles has an $H^1(\mathbf{Z}/2) \simeq \mathbf{Z}/2$. After ramification over the base S and étale localization at the singular point, there are therefore two possible cases. They are covered by cases I and II of 2.3.

Yours,

P. Deligne